class09: Halloween Candy Mini-Project

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Background

In today's mini-project we will analyze candy data with ggplot, basic statistics, correlation analysis, and principal component analysis methods we have been learning thus far.

Data Import

This data comes as a CSV file from the 538 website:

```
candy_file <- "candy-data.csv"

candy = read.csv(candy_file, row.names=1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
```

```
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

There are 85 different candy types in this dataset.

Q2. How many fruity candy types are in the dataset?

```
sum(candy[,"fruity"])
```

```
[1] 38
```

There are 38 fruity candy types in the candy dataset.

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Almond Joy",]$winpercent
```

```
[1] 50.34755
```

My favorite candy of Almond Joy in this dataset has a win percent of 50.34755.

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat",]$winpercent
```

```
[1] 76.7686
```

My favorite candy of Kit Kat in this dataset has a win percent of 76.7686.

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```r
candy["Tootsie Roll Snack Bars",]$winpercent
```

```
[1] 49.6535
```

My favorite candy of Kit Kat in this dataset has a win percent of 49.6535.

```r
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

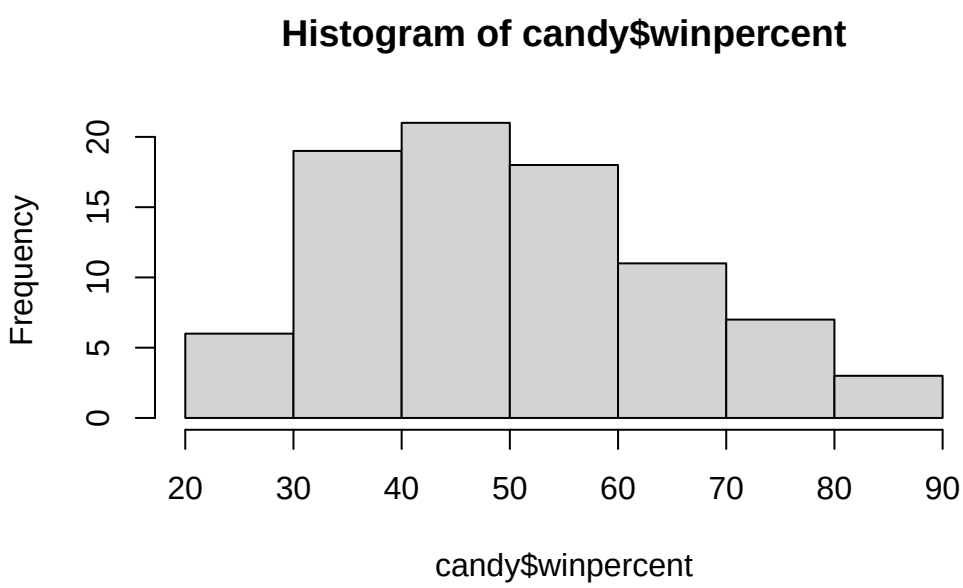
The `winpercent` column looks to be on a different scale than the majority of the other columns in the dataset.

> Q7. What do you think a zero and one represent for the candy$chocolate column?

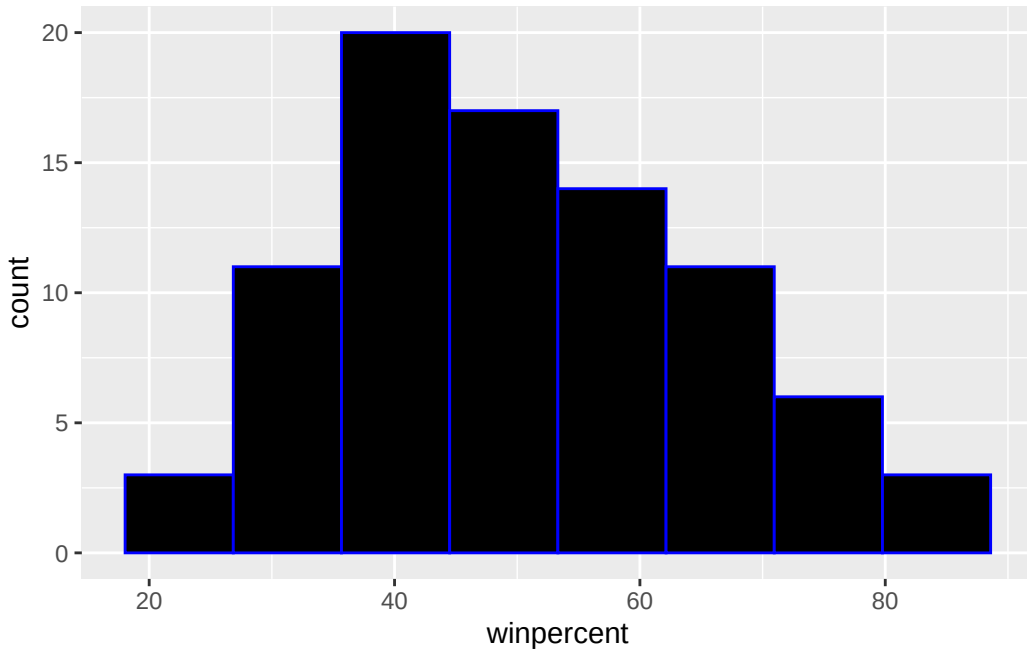
I think a zero in the `chocolate` column of the `candy` dataset means that a particular candy does not have chocolate in it and then a one in the `chocolate` column of the `candy` dataset means that a particular candy does have chocolate in it.

> Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$"winpercent", breaks=8)
```




```
library(ggplot2)
ggplot(candy, aes(x = winpercent)) + geom_histogram(bins=8, fill='black',col='blue')
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values is not symmetrical since has a right tail.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

The center of the distribution is above 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
chocolate <- candy[candy$chocolate == 1,]
fruity <- candy[candy$fruity == 1,]
print(paste("Fruity candy mean win percentage:",mean(fruity$winpercent)))
```

```
[1] "Fruity candy mean win percentage: 44.1197414210526"
```

```r
print(paste("Chocolate candy mean win percentage:",mean(chocolate$winpercent)))
```

```
[1] "Chocolate candy mean win percentage: 60.9215294054054"
```

On average, chocolate candy has a higher ranking with respect to win percentage.

Q12. Is this difference statistically significant?

```r
t.test(chocolate$winpercent, fruity$winpercent)
```

```
    Welch Two Sample t-test

data:  chocolate$winpercent and fruity$winpercent
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

This difference is statistically significant since when doing a t-test on these two samples of the larger candy dataset, the obtained p-vlaue is very low and therefore we can reject the null hypothesis that there is no statistical difference between win percentages of chocolate and fruity candies.

Q13. What are the five least liked candy types in this set?

```r
inds <- order(candy$winpercent)
head(candy[inds,], n=5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
```

```
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

The five least liked candy types in this dataset are Nik L Nip, Boston Baked Beans, Chiclets, Supper Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

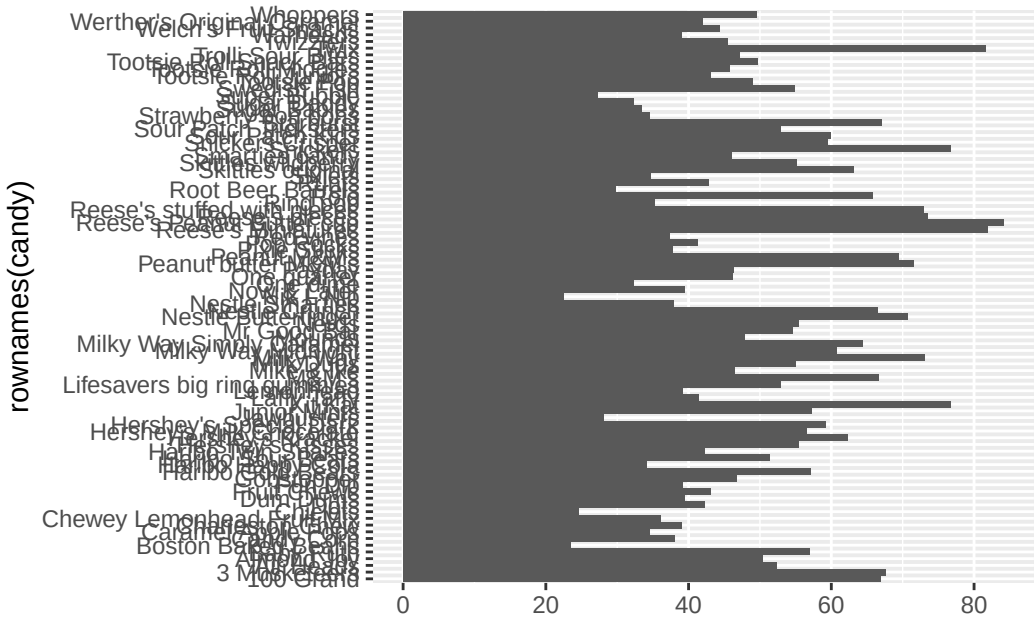
```r
head(candy[order(candy$winpercent, decreasing=TRUE),], n=5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Reese's Peanut Butter cup         1      0       0              1      0
Reese's Miniatures                1      0       0              1      0
Twix                              1      0       1              0      0
Kit Kat                           1      0       0              0      0
Snickers                          1      0       1              1      1
                          crispedricewafer hard bar pluribus sugarpercent
Reese's Peanut Butter cup                0    0   0        0        0.720
Reese's Miniatures                       0    0   0        0        0.034
Twix                                     1    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Snickers                                 0    0   1        0        0.546
                          pricepercent winpercent
Reese's Peanut Butter cup        0.651   84.18029
Reese's Miniatures               0.279   81.86626
Twix                             0.906   81.64291
Kit Kat                          0.511   76.76860
Snickers                         0.651   76.67378
```

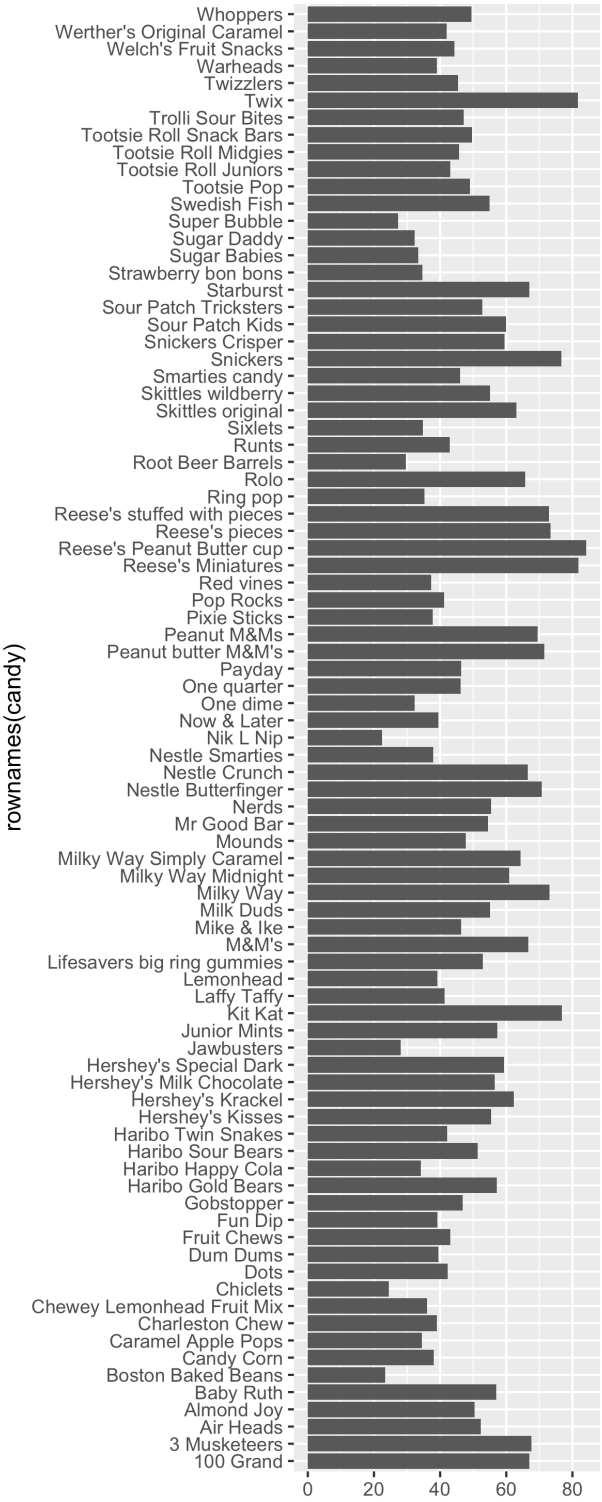
The top 5 all time favorite candy types out of this dataset are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

```r
ggplot(candy, aes(x = rownames(candy), y = winpercent)) +
  geom_bar(stat = "identity") +
  coord_flip() + ylab("")
```



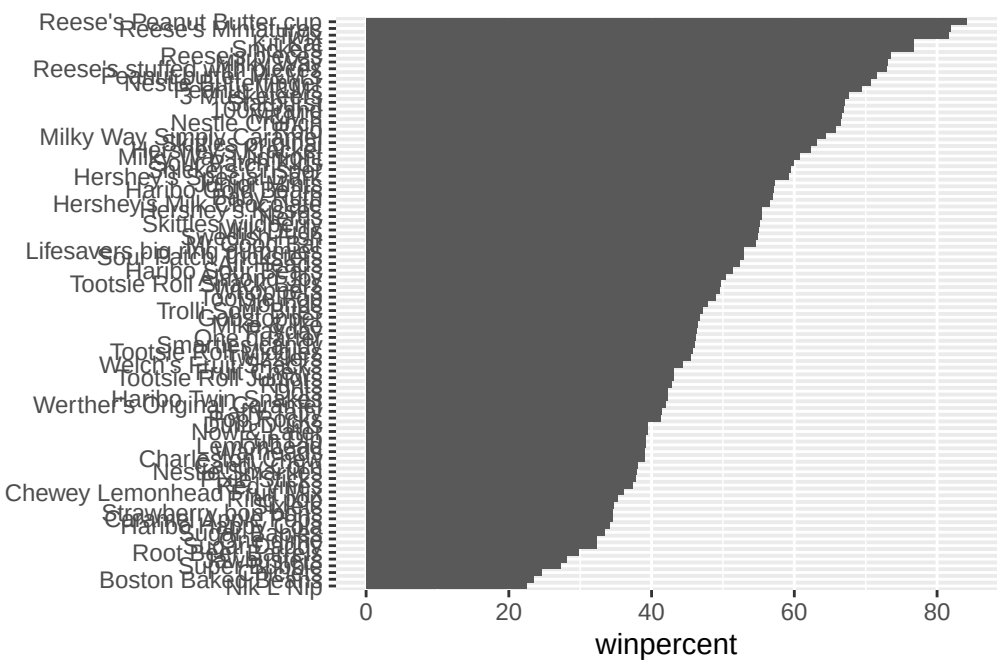

```r
ggsave("barplot.png", height=10, width=4)
```


rownames(candy)
Whoppers
Werther's Original Caramel
Welch's Fruit Snacks
Warheads
Twizzlers
Twix
Trolli Sour Bites
Tootsie Roll Snack Bars
Tootsie Roll Midgies
Tootsie Roll Juniors
Tootsie Pop
Swedish Fish
Super Bubble
Sugar Daddy
Sugar Babies
Strawberry bon bons
Starburst
Sour Patch Tricksters
Sour Patch Kids
Snickers Crisper
Snickers
Smarties candy
Skittles wildberry
Skittles original
Sixlets
Runts
Root Beer Barrels
Rolo
Ring pop
Reese's stuffed with pieces
Reese's pieces
Reese's Peanut Butter cup
Reese's Miniatures
Red vines
Pop Rocks
Pixie Sticks
Peanut M&Ms
Peanut butter M&M's
Payday
One quarter
One dime
Now & Later
Nik L Nip
Nestle Smarties
Nestle Crunch
Nestle Butterfinger
Nerds
Mr Good Bar
Mounds
Milky Way Simply Caramel
Milky Way Midnight
Milky Way
Milk Duds
Mike & Ike
M&M's
Lifesavers big ring gummies
Lemonhead
Laffy Taffy
Kit Kat
Junior Mints
Jawbusters
Hershey's Special Dark
Hershey's Milk Chocolate
Hershey's Krackel
Hershey's Kisses
Haribo Twin Snakes
Haribo Sour Bears
Haribo Happy Cola
Haribo Gold Bears
Gobstopper
Fun Dip
Fruit Chews
Dum Dums
Dots
Chiclets
Chewey Lemonhead Fruit Mix
Charleston Chew
Caramel Apple Pops
Candy Corn
Boston Baked Beans
Baby Ruth
Almond Joy
Air Heads
3 Musketeers
100 Grand
0
20
40
60
80

Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
ggplot(candy, aes(winpercent, reorder(rownames(candy),winpercent)), y = winpercent) +
  geom_bar(stat = "identity") + ylab("")
```

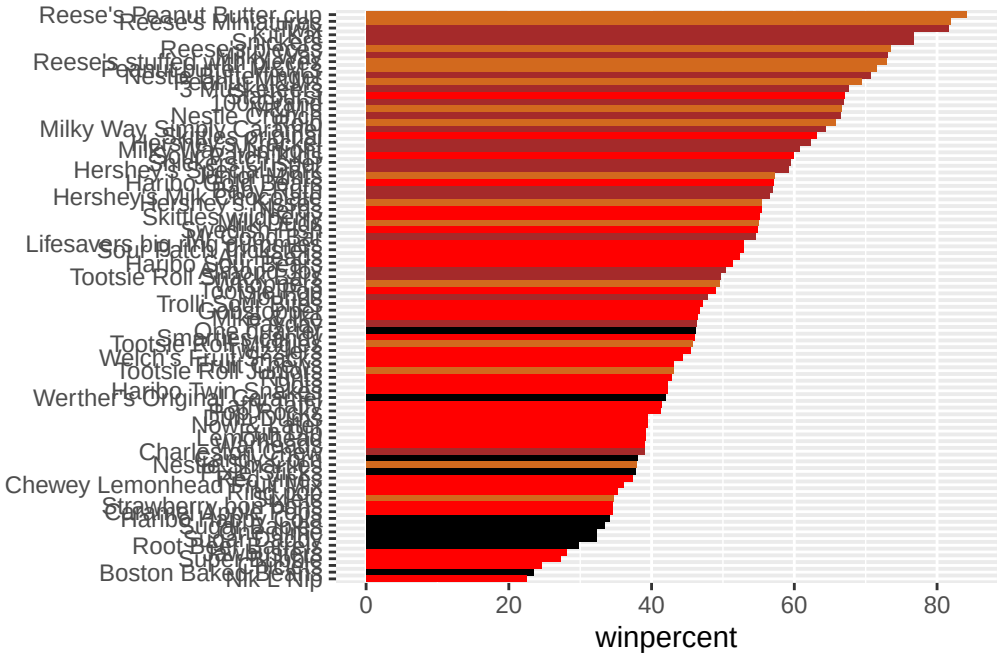
```
Warning in fortify(data, ...): Arguments in `...` must be used.
x Problematic argument:
* y = winpercent
i Did you misspell an argument name?
```



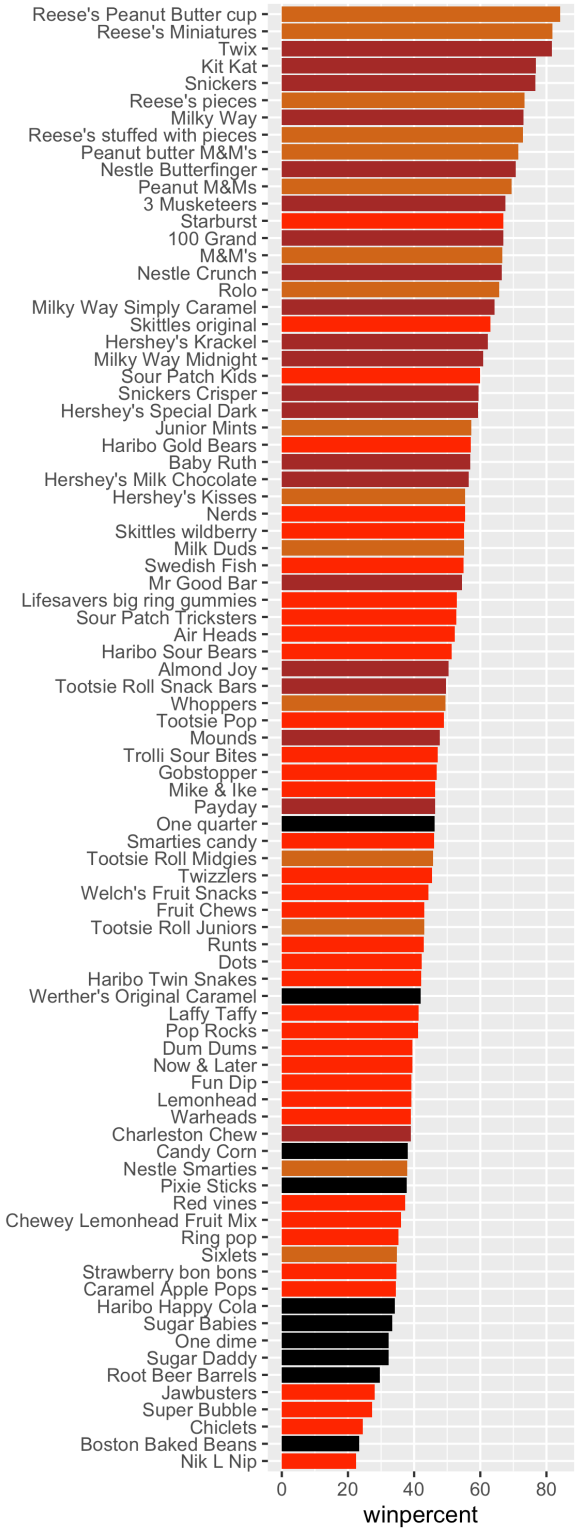
Let's add some color to this plot

```
my_cols=rep("black", nrow(candy))
my_cols[candy$chocolate ==1] = "chocolate"
my_cols[candy$bar == 1] = "brown"
my_cols[candy$fruity == 1] = "red"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)+ ylab("")
```




```
ggsave("barplot_colors.png", height=10, width=4)
```


Reese's Peanut Butter cup
Reese's Miniatures
Twix
Kit Kat
Snickers
Reese's pieces
Milky Way
Reese's stuffed with pieces
Peanut butter M&M's
Nestle Butterfinger
Peanut M&Ms
3 Musketeers
Starburst
100 Grand
M&M's
Nestle Crunch
Rolo
Milky Way Simply Caramel
Skittles original
Hershey's Krackel
Milky Way Midnight
Sour Patch Kids
Snickers Crisper
Hershey's Special Dark
Junior Mints
Haribo Gold Bears
Baby Ruth
Hershey's Milk Chocolate
Hershey's Kisses
Nerds
Skittles wildberry
Milk Duds
Swedish Fish
Mr Good Bar
Lifesavers big ring gummies
Sour Patch Tricksters
Air Heads
Haribo Sour Bears
Almond Joy
Tootsie Roll Snack Bars
Whoppers
Tootsie Pop
Mounds
Trolli Sour Bites
Gobstopper
Mike & Ike
Payday
One quarter
Smarties candy
Tootsie Roll Midgies
Twizzlers
Welch's Fruit Snacks
Fruit Chews
Tootsie Roll Juniors
Runts
Dots
Haribo Twin Snakes
Werther's Original Caramel
Laffy Taffy
Pop Rocks
Dum Dums
Now & Later
Fun Dip
Lemonhead
Warheads
Charleston Chew
Candy Corn
Nestle Smarties
Pixie Sticks
Red vines
Chewey Lemonhead Fruit Mix
Ring pop
Sixlets
Strawberry bon bons
Caramel Apple Pops
Haribo Happy Cola
Sugar Babies
One dime
Sugar Daddy
Root Beer Barrels
Jawbusters
Super Bubble
Chiclets
Boston Baked Beans
Nik L Nip
0
20
40
60
80
winpercent

Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Sixlets.

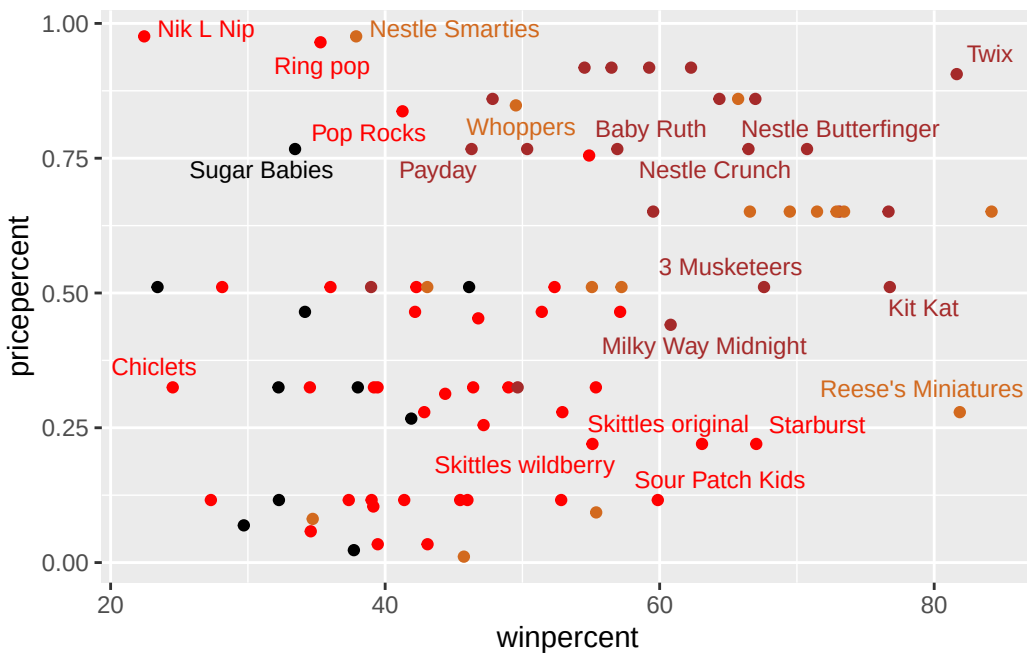
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburts.

```
library(ggrepel)

ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```

```
Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
bar <- candy[candy$bar == 1,]
print(paste("Winpercentage divided by pricepercentage for chocolate candies",mean(chocolate$w
```

```
[1] "Winpercentage divided by pricepercentage for chocolate candies 96.3700973169238"
```

```
print(paste("Winpercentage divided by pricepercentage for fruity candies",mean(fruity$winperc
```

```
[1] "Winpercentage divided by pricepercentage for fruity candies 132.596503661683"
```

```
print(paste("Winpercentage divided by pricepercentage for bar candies",mean(bar$winpercent)/m
```

```
[1] "Winpercentage divided by pricepercentage for bar candies 84.4344807536626"
```

Fruity candies are the highest ranked in terms of winpercent for the least money.

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
vec <- ord[1:5]
win_check <- 10000
v <- c()
for (i in rownames(candy)[vec]) {
  if (candy[i, "winpercent"] < win_check){
    win_check = candy[i, "winpercent"]
    v <- c(v, i)
  }
}
v[length(v)]
```

```
[1] "Nik L Nip"
```

Of the top 5 most expensive candy types in the dataset, the least popular is "Nik L Nip".

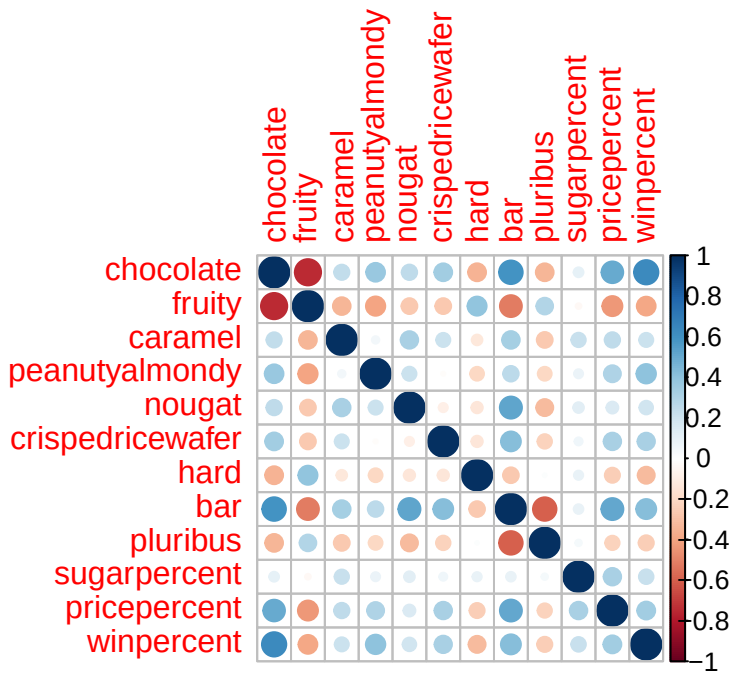
Exploring the correlation structure

Pearson Correlation values range from -1 to +1

```r
library(corrplot)
```

```
corrplot 0.95 loaded
```

```r
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity candies and chocolate candies are anti-correlated as evidenced by the red dot in their intersection in the plot above.

Q23. Similarly, what two variables are most positively correlated?

Chocolate candies and winpercent are most positively correlated as evidenced by the dark blue dot in their intersection in the plot above.

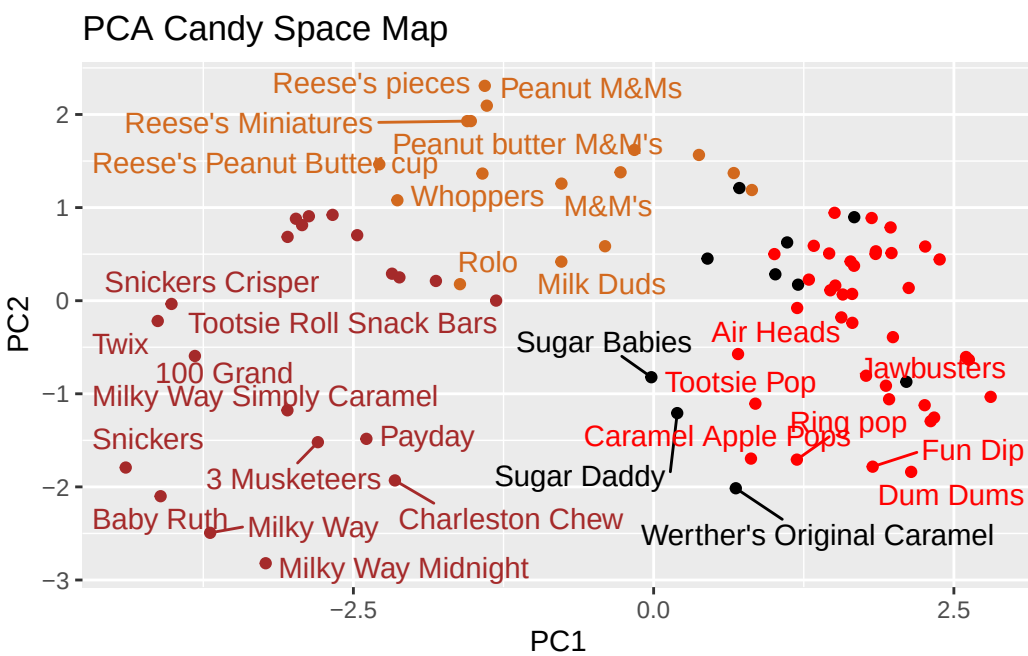
```r
pca<-prcomp(candy, scale=T)
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

The main results figure: the PCA score plot:

```
p<-ggplot(pca$x)+aes(PC1,PC2, label = rownames(pca$x)) + geom_point(col=my_cols) + geom_text_
p
```

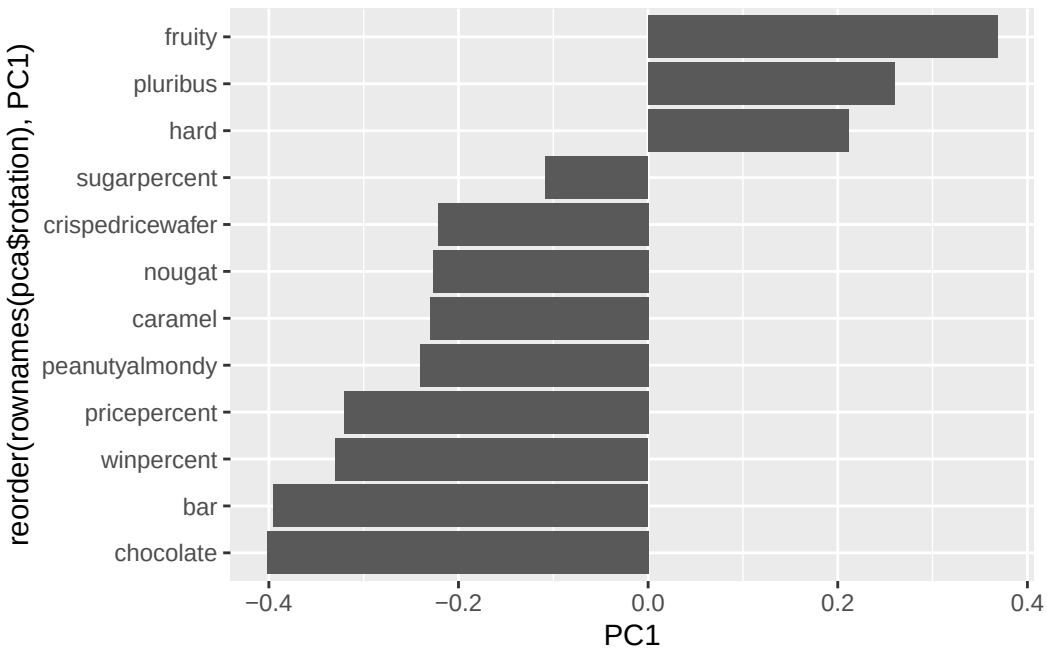
```
Warning: ggrepel: 54 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```




```
#library(plotly)
#ggplotly(p)
```

The "loadings" plot for PC1

```r
ggplot(pca$rotation) +
  aes(x = PC1,
      y = reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```



> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Chocolate, bar, and winpercent are all picked up strongly by PC1 in the positive direction. These make sense to me. We saw this relationship before in the correlation matrix.

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

The "winning candy" is made of a combination of chocolate, winpercent, bar, and pricepercent. The visualization, correlation, and PCA all compliment each other by presenting the characterisitcs of chocolate, winpercent, bar, and pricepercent as larger components to make a winning candy.